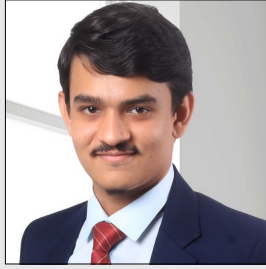


Dr. Parth Nayak

Data Scientist | AI Researcher | Astrophysicist



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 in [dr-parth-nayak-astro](#)
 🐙 [par-nay](#)

EDUCATION

🎓 **Ph.D. in Astrophysics** (Dr. rer. nat.)
 LMU Munich | Dec 2025
 🎓 **B.Sc.+M.Sc. in Physics** (dual degree)
 IISER Bhopal | Jul 2020

SKILLS

ML / Deep Learning

Neural networks (CNNs, MLPs, Transformers), simulation-based inference (SBI), generative models (normalizing flows, diffusion models, autoencoders, GANs)

Other AI

Gaussian processes, Bayesian & Monte Carlo methods, mixture models, transfer learning, domain adaptation, evolutionary algorithms

Programming

Python, LaTeX (proficient); HTML, CSS, Javascript (intermediate); C++, Java (basic)

Python packages

TensorFlow, Keras, Optuna, NumPy, SciPy, Pandas, Matplotlib, H5Py, AstroPy, HealPy, *among others*

Other IT

High-performance computing (HPC), Git version control, LLM prompt engineering, Markdown, Mac/MS Office, Shell (bash, Z shell)

LANGUAGES

English (full working proficiency); Gujarati, Hindi (native/bilingual); German (B1/intermediate); French (beginner)

HIGHLIGHTS

- 5+ years of research experience in Data Science and AI algorithm development
- Authored 4 research papers on novel AI applications in astrophysics
- Delivered 10+ technical talks at international conferences and research institutes
- Received 5+ academic awards and fellowships

RELEVANT EXPERIENCE

Doctoral Candidate: Astrophysics & Data Science LMU Munich | Oct 2021 - Dec 2025
Hightech Agenda Bayern chair of Astrophysics, Cosmology & AI + Munich Center for Machine Learning (MCML)

- Developed a ML inference framework “*LyaNNA*” [[🐙 par-nay/lyanna](#)] based on a novel 1D ResNet with Python/TensorFlow from scratch that **outperforms conventional methods** by >100% in inference precision for complex, realistic, noisy spectroscopic data (simulation-based inference)
- Developed “*SynTH*” [[🐙 par-nay/synth](#)] to model intergalactic absorption from massive cosmological **hydrodynamic simulations** and synthesize training data for *LyaNNA*
- Authored 3 research papers on AI for inference in astrophysics [[✍Nayak+ 2024, 2025](#), [Chang+ 2025](#)]

Master's Thesis: Astrophysics & Data Science IISER Bhopal | Aug 2019 - Jul 2020

- Developed a Genetic Algorithm framework “*GA-ILC*” to optimize the reconstruction of cosmic microwave background (CMB) from contaminated observations
 - ▶ A novel, **scalable** evolutionary algorithm [[🐙 par-nay/genepy](#)] to extract faint cosmological signals from high-noise environments (foreground removal)
- Authored one research paper on evolutionary algorithms in astrophysics [[✍Nayak & Saha 2022](#)]

LEADERSHIP & SUPERVISION

- Co-supervisor of LMU Master's theses and labs Oct 2021 - Dec 2025
 Designed and supervised two LMU Master's theses, and co-supervised 5+ computational and instrumental lab courses
- Co-organizer of “Astronomy on Tap” (AoT) Munich since Oct 2025
 Coordinating the monthly outreach event to engage the general public with accessible scientific talks and pub-quizzes
- Co-organizer of “Meet the Speaker” @LMU Observatory Oct 2023 - May 2025
 Revitalized the weekly series by relocating it to the historic 200-year old telescope dome, a unique atmosphere for students for informal networking with international experts
- Coordinator & Senior Advisor of IISERB Astronomy Club (IBAC) 2016 - 2020
 Played a foundational role as the club's second coordinator, establishing enduring traditions and growing the community into one of India's most popular student astronomy clubs
- Physics Department Representative Aug 2017 - Jul 2018
 Unanimously elected to serve as the primary liaison between faculty and a class of 60+ physics major students

SELECTED AWARDS & FELLOWSHIPS

- DAAD Research Grant — Ph.D. scholarship Germany | Oct 2021 - Dec 2025
- Proficiency Medal in Physics — IISER Bhopal India | Jul 2020
- DAAD-WISE Scholarship Germany | May - Jul 2019
- INSPIRE Scholarship for Higher Education India | Aug 2015 - Jul 2020